

Sample questions for CO1 – Assessment Tool 3 – Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.
6. What are web services? Explain how web services facilitate communication between cloud applications.
7. Discuss the major advantages and challenges of cloud computing. Explain how organizations can benefit from cloud adoption.

1. cloud computing:-

cloud computing is the process of Transfers the data and it has four types of cloud computing that make it different from Traditional on premise computing.

1. On - demand self service :

Users can access computing resource whenever needed that human interaction without human interaction.

2. Board network access:-

Services can be accessed from anywhere using devices like laptop, mobile phones, and tablet.

3. Resource pooling:- Resource are shared among multiple users through virtualiza

4. Rapid elasticity:-

Resource can be increases or decreases quickly based on demand

So, this are the cloud computing and four characteristics.

2.

evolution of cloud computing

1. Hardware evolution:-

Large mainframe computers were used for centralized computing.

2. Personal computers:-

computers becomes affordable and allowing individual use.

3. Network:-

LAN or WAN enabled communication and resource sharing.

4. Virtualization:-

one physical server could run multiple virtual machines, improving resource utilization.

5. Internet and web:-

High speed Internet enable online software and services.

6. cloud computing:-

cloud computation is used to process and Transfers the data.

So, the evolution of cloud computing from hardware evolution to internet software evolution.

server Virtualization:-

It divides one physical server into multiple using hypervisor. each virtual machines works independently and with its own operating system.

Advantages:-

1. Better hardware virtual utilization.
2. Lower cost
3. Easy management
4. Fast deployment.

Differentiate between Virtualization and cloud computing.

Virtualization	cloud computing.
Technology that creates virtual machines	cloud service models that delivers the resources.
It focus on the development and the deployment, and it also runs on physical server.	It uses Virtual to privacy server.
Focuses on efficient resources	It focuses on delivering resources.
It is used based on Virtualization techniques.	It is used based on cloud computing techniques.

4. data centers:-

Data is a facilitate the resource centers that contains servers, storage, devices, networking, equipment, cooling systems and power supplies. It is called as data centers.

Roles:-

1. stores and management.
2. Ensures the high availability and Back up and resource disaster recovery.
3. It host the application and websites.
4. protects the data through firewalls.

Advantages:-

1. Easy to maintain and manage.
2. provides the high availability backups and disaster recovery.
3. It host the application and websites.
4. It protects the datas through the firewalls.
5. It is fast deployment.

5.

Services	Description	Example
IaaS Infrastructure as a Services.	It requires to stores and Services	Amazon AWS
PaaS Platform as a Services	It requires platform to develop and deployment in the servers.	Google
SaaS Selfservice as a Services	It requires the Self services that receives through the Internet.	google Docs
XaaS Anything as a Services.	It requires the Anything services from the internet	Microsoft Azure

Advantages:-

IaaS → flexible, easy to manage, and stores the data.

PaaS → low cost, easy to deploy and development.

SaaS → high or low cost, easy to define

XaaS → defines the origin, easy to maintain.